

CH61014: Advanced Heat Transfer

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Sturm – Liouville Problems
Ordinary Fourier Series
Complete Fourier Series
Fourier Bessel Series

Sturm – Liouville Problems

Consider the following characteristic-value problem, composed of the linear and homogeneous second-order differential equation of the general form

$$\frac{d}{dx} \left[p(x) \frac{dy}{dx} \right] + [q(x) + \lambda w(x)] y = 0$$

and the two homogeneous linear boundary condition prescribed at the ends of the finite interval (a, b) .

$$\alpha_1 y(a) + \beta_1 \frac{dy(a)}{dx} = 0, \quad \alpha_1^2 + \beta_1^2 > 0$$

$$\alpha_2 y(b) + \beta_2 \frac{dy(b)}{dx} = 0, \quad \alpha_2^2 + \beta_2^2 > 0$$

Note:

1. The functions $p(x)$, $q(x)$ and $w(x)$ are *real valued* and *continuous* (including dp/dx).
2. Also, $p(x)$ and $w(x)$ are *positive* over the entire interval (a, b) , including the end points, while $q(x) \leq 0$ in the same interval.
3. Furthermore, α_1 , α_2 , β_1 and β_2 are given real constants, and λ is an unspecified parameter, independent of x .

Characteristic-value problems of this type are known as *Sturm–Liouville problems*.

Sturm – Liouville Problems

$$\frac{d}{dx} \left[p(x) \frac{dy}{dx} \right] + [q(x) + \lambda w(x)] y = 0 \quad \alpha_1 y(a) + \beta_1 \frac{dy(a)}{dx} = 0, \quad \alpha_1^2 + \beta_1^2 > 0$$
$$\alpha_2 y(b) + \beta_2 \frac{dy(b)}{dx} = 0, \quad \alpha_2^2 + \beta_2^2 > 0$$

Nontrivial solutions of the above problem exist, in general, for a particular set of values $\lambda_1, \lambda_2, \dots, \lambda_n$ of the parameter λ . These are the characteristic values, (or the eigenvalues), and the corresponding solutions are the characteristic functions (or the eigenfunctions) of the problem.

Let λ_m and λ_n be any two distinct characteristic values, and $\phi_m(x)$ and $\phi_n(x)$ be the corresponding characteristic functions, respectively. These functions satisfy the above differential equation. It can be shown:

$$(\lambda_n - \lambda_m) \int_a^b \phi_m(x) \phi_n(x) w(x) dx = 0 \quad \rightarrow \quad \int_a^b \phi_m(x) \phi_n(x) w(x) dx = 0, \quad m \neq n$$

Thus, the characteristic functions of the Sturm–Liouville problem form an orthogonal set with respect to the weight function $w(x)$ on the interval (a, b) .

Sturm – Liouville Problems: Singular End Points

$$\frac{d}{dx} \left[p(x) \frac{dy}{dx} \right] + [q(x) + \lambda w(x)] y = 0 \quad \alpha_1 y(a) + \beta_1 \frac{dy(a)}{dx} = 0, \quad \alpha_1^2 + \beta_1^2 > 0$$
$$\alpha_2 y(b) + \beta_2 \frac{dy(b)}{dx} = 0, \quad \alpha_2^2 + \beta_2^2 > 0$$

If it so happens that $p(a) = 0$, then the characteristic functions corresponding to different characteristic values will be orthogonal with respect to the weight function $w(x)$ on (a, b) , provided that $y(x)$ and dy/dx are both finite at $x = a$.

In this case, the boundary condition at $x = a$ is replaced by the requirement that $y(x)$ and dy/dx be finite at $x = a$ when $p(a) = 0$.

Similarly, if $p(b) = 0$, we can then replace the second boundary condition (at $x = b$) by the requirement that $y(x)$ and dy/dx be finite at $x = b$.

If both $p(a) = 0$ and $p(b) = 0$, then neither boundary condition (at $x = a$ and $x = b$) as specified above is needed to ensure the orthogonality of the characteristic functions, provided that both $y(x)$ and dy/dx are finite at $x = a$ and $x = b$.

Sturm – Liouville Problems: Periodic BC

If $p(a) = p(b)$, then the boundary conditions can be replaced by

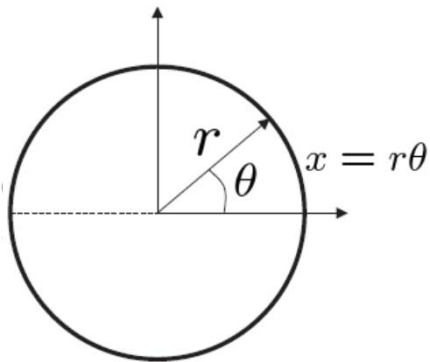
$$y(a) = y(b)$$

$$\frac{dy(a)}{dx} = \frac{dy(b)}{dx}$$

These are called *periodic boundary conditions* and they are satisfied, in particular, if the solution $y(x)$ is required to be periodic, of period $(b - a)$.

Heat Conduction in a Circular Ring:

We consider a thin wire of length $2L$ which is bent into the shape of a circle. Then at the endpoints we must match the temperature and the flux.



$$\begin{aligned}\nabla^2 u &= \frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} \\ &= \frac{\partial^2 u}{\partial (r\theta)^2}\end{aligned}$$

$$\text{let } x = r\theta \text{ then } \frac{\partial^2 u}{\partial (r\theta)^2} = u_{xx}.$$

$$\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}, \quad -L < x < L$$

$$u(-L, t) = u(L, t), \quad t > 0$$

$$\frac{\partial u}{\partial x}(-L, t) = \frac{\partial u}{\partial x}(L, t), \quad t > 0$$

$$u(x, 0) = f(x), \quad -L \leq x \leq L$$

Thin circular wire - no radial temperature dependence. $u(r, \theta) = u(\theta)$

Sturm – Liouville Problems

$$\frac{d}{dx} \left[p(x) \frac{dy}{dx} \right] + [q(x) + \lambda w(x)] y = 0$$

$$\alpha_1 y(a) + \beta_1 \frac{dy(a)}{dx} = 0, \quad \alpha_1^2 + \beta_1^2 > 0$$

$$\alpha_2 y(b) + \beta_2 \frac{dy(b)}{dx} = 0, \quad \alpha_2^2 + \beta_2^2 > 0$$

Note:

The characteristic values of the Sturm–Liouville problem are all *real* and *nonnegative*, and the corresponding characteristic functions are *real*. The parameter λ can, therefore, be replaced by λ^2 with no loss in the generality of the problem.

In addition, there is *only one* characteristic function $\phi_n(x)$ which corresponds to each characteristic value λ_n , except when the periodic boundary conditions are involved.

More importantly, the characteristic functions form a *complete* orthogonal set.

Generalized Fourier Series

Consider a set of functions $\{\phi_n(x); n = 0, 1, 2, \dots\}$ orthogonal with respect to a weight function $w(x)$ on the finite interval (a, b) .

Expand an arbitrary function $f(x)$ in a series of these functions as

$$f(x) = A_0\phi_0(x) + A_1\phi_1(x) + \dots + A_n\phi_n(x) + \dots$$

$$\rightarrow f(x) = \sum_{n=0}^{\infty} A_n\phi_n(x)$$

Note:
Coefficients are unknown

Now evaluate unknown coefficients:

$$\int_a^b f(x)\phi_n(x)w(x)dx = \int_a^b \left[\sum_{k=0}^{\infty} A_k\phi_k(x) \right] \phi_n(x)w(x)dx$$

Use orthogonal property: $\int_a^b f(x)\phi_n(x)w(x)dx = A_n \int_a^b [\phi_n(x)]^2 w(x)dx$

$$A_n = \frac{1}{N_n} \int_a^b f(x)\phi_n(x)w(x)dx \quad N_n = \int_a^b [\phi_n(x)]^2 w(x)dx$$

Ordinary Fourier Series

The *ordinary Fourier series*, or simply the *Fourier series*, are developed from the characteristic functions of the following characteristic-value problem for different combinations of the boundary conditions:

$$\frac{d^2 y}{dx^2} + \lambda^2 y(x) = 0$$

$$\frac{d}{dx} \left[p(x) \frac{dy}{dx} \right] + [q(x) + \lambda w(x)] y = 0$$

$$\alpha_1 y(0) + \beta_1 \frac{dy(0)}{dx} = 0, \quad \alpha_1^2 + \beta_1^2 > 0$$

This characteristic-value problem is a special case of the Sturm–Liouville system with $p(x) = 1$, $q(x) = 0$, $w(x) = 1$, and λ replaced by λ^2 .

$$\alpha_2 y(L) + \beta_2 \frac{dy(L)}{dx} = 0, \quad \alpha_2^2 + \beta_2^2 > 0$$

Let the functions $\phi_n(x)$ are the characteristic functions (orthogonal) of the characteristic-value problem.

The *ordinary Fourier series*, or simply a *Fourier series*, of $f(x)$ on the interval $(0, L)$:

$$f(x) = \sum_{n=0}^{\infty} A_n \phi_n(x), \quad 0 < x < L \quad A_n = \frac{1}{N_n} \int_0^L f(x) \phi_n(x) dx \quad N_n = \int_0^L [\phi_n(x)]^2 dx$$

Ordinary Fourier Series: How Many?

$$\frac{d^2 y}{dx^2} + \lambda^2 y(x) = 0$$

$$\alpha_1 y(0) + \beta_1 \frac{dy(0)}{dx} = 0, \quad \alpha_1^2 + \beta_1^2 > 0$$

$$\alpha_2 y(L) + \beta_2 \frac{dy(L)}{dx} = 0, \quad \alpha_2^2 + \beta_2^2 > 0$$

There are nine different combinations of the boundary conditions. Accordingly, corresponding to each combination there will be a series expansion of the form given by:

$$f(x) = \sum_{n=0}^{\infty} A_n \phi_n(x), \quad 0 < x < L$$

We will discuss the two most common cases:

- Fourier Sine Series
- Fourier Cosine Series

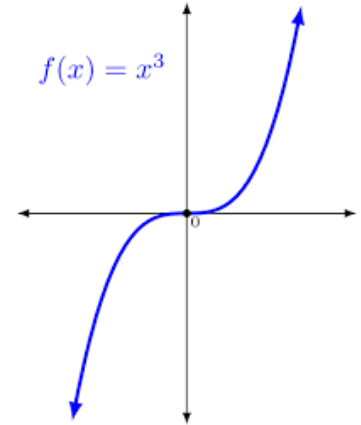
Fourier Sine Series

Consider the following characteristic-value problem:

$$\frac{d^2 y}{dx^2} + \lambda^2 y = 0$$

$$y(0) = 0 \quad \text{and} \quad y(L) = 0$$

A function is "odd" when:
 $-f(x) = f(-x)$ for all x



Characteristic functions and characteristic values:

$$\phi_n(x) = \sin \lambda_n \quad \text{and} \quad \lambda_n = \frac{n\pi}{L}, \quad n = 1, 2, 3, \dots$$

$$f(x) = \sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{L} x, \quad 0 < x < L \quad \text{Eq. (A)}$$

**Fourier sine series
 representation of $f(x)$
 on the interval $(0, L)$**

$$A_n = \frac{1}{N_n} \int_0^L f(x) \sin \frac{n\pi}{L} x \, dx$$

$$N_n = \int_0^L \sin^2 \frac{n\pi}{L} x \, dx = \frac{L}{2}$$

Note that the RHS of **Eq. (A)** is a periodic function of period $2L$, and also is an odd function of x . Therefore, if **Eq. (A)** converges to $f(x)$ in $(0, L)$, it will also converge to $-f(-x)$ in $(-L, 0)$. In other words, if $f(x)$ is an odd function of x , then **Eq. (A)** will represent $f(x)$ not only in $(0, L)$, but also in $(-L, L)$.

Fourier Cosine Series

Consider the following characteristic-value problem:

$$\frac{d^2 y}{dx^2} + \lambda^2 y = 0$$

$$\frac{dy(0)}{dx} = 0$$

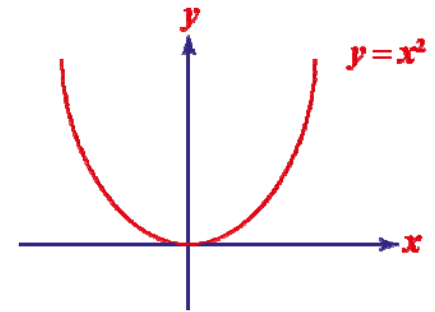
and

$$\frac{dy(L)}{dx} = 0$$

Characteristic functions and characteristic values:

$$\phi_n(x) = \cos \lambda_n x \quad \text{and} \quad \lambda_n = \frac{n\pi}{L}, \quad n = 0, 1, 2, \dots$$

A function is "even" when:
 $f(x) = f(-x)$ for all x



$$f(x) = \sum_{n=0}^{\infty} A_n \cos \frac{n\pi}{L} x = A_0 + \sum_{n=1}^{\infty} A_n \cos \frac{n\pi}{L} x, \quad 0 < x < L \quad \text{Eq. (B)}$$

*Fourier cosine series
 representation of $f(x)$
 on the interval $(0, L)$*

$$A_n = \frac{1}{N_n} \int_0^L f(x) \cos \frac{n\pi}{L} x dx \quad N_n = \int_0^L \cos^2 \frac{n\pi}{L} x dx = \begin{cases} L, & n = 0 \\ \frac{L}{2}, & n = 1, 2, 3, \dots \end{cases}$$

Note that the RHS of **Eq. (B)** is a periodic function of period $2L$, and also is an even function of x . Therefore, if **Eq. (B)** converges to $f(x)$ in $(0, L)$, it will also converge to $f(-x)$ in $(-L, 0)$. In other words, if $f(x)$ is an even function of x , then **Eq. (B)** will represent $f(x)$ not only in $(0, L)$, but also in $(-L, L)$.

Complete Fourier Series

Any function of x , say $F(x)$, can be written as:

$$F(x) = \frac{1}{2}[F(x) - F(-x)] + \frac{1}{2}[F(x) + F(-x)]$$

Odd function

Even function

If $f(x)$ is odd:

$$f(-x) = -f(x)$$

If $f(x)$ is even:

$$g(-x) = g(x)$$

$$F(x) = \sum_{n=1}^{\infty} a_n \sin \frac{n\pi}{L} x + \sum_{n=0}^{\infty} b_n \cos \frac{n\pi}{L} x, \quad -L < x < L$$

or,

$$F(x) = b_0 + \sum_{n=1}^{\infty} \left[a_n \sin \frac{n\pi}{L} x + b_n \cos \frac{n\pi}{L} x \right], \quad -L < x < L$$

$$\begin{aligned} a_n &= \frac{2}{L} \int_0^L \left\{ \frac{1}{2} [F(x) - F(-x)] \right\} \sin \frac{n\pi}{L} x dx \\ &= \frac{1}{L} \int_{-L}^L F(x) \sin \frac{n\pi}{L} x dx, \quad n = 1, 2, 3, \dots \end{aligned}$$

$$b_0 = \frac{1}{L} \int_0^L \left\{ \frac{1}{2} [F(x) + F(-x)] \right\} dx = \frac{1}{2L} \int_{-L}^L F(x) dx$$

$$\begin{aligned} b_n &= \frac{2}{L} \int_0^L \left\{ \frac{1}{2} [F(x) + F(-x)] \right\} \cos \frac{n\pi}{L} x dx \\ &= \frac{1}{L} \int_{-L}^L F(x) \cos \frac{n\pi}{L} x dx, \quad n = 1, 2, 3, \dots \end{aligned}$$

Complete Fourier Series

$$\begin{aligned}a_n &= \frac{2}{L} \int_0^L \left\{ \frac{1}{2} [F(x) - F(-x)] \right\} \sin \frac{n\pi}{L} x dx \\&= \frac{1}{L} \int_{-L}^L F(x) \sin \frac{n\pi}{L} x dx, \quad n = 1, 2, 3, \dots\end{aligned}$$

$$b_0 = \frac{1}{L} \int_0^L \left\{ \frac{1}{2} [F(x) + F(-x)] \right\} dx = \frac{1}{2L} \int_{-L}^L F(x) dx$$

$$\begin{aligned}b_n &= \frac{2}{L} \int_0^L \left\{ \frac{1}{2} [F(x) + F(-x)] \right\} \cos \frac{n\pi}{L} x dx \\&= \frac{1}{L} \int_{-L}^L F(x) \cos \frac{n\pi}{L} x dx, \quad n = 1, 2, 3, \dots\end{aligned}$$

If $F(x)$ is a periodic function with period $2L$, then the coefficients a_n and b_n can be determined equivalently from

$$a_n = \frac{1}{L} \int_c^{c+2L} F(x) \sin \frac{n\pi}{L} x dx, \quad n = 1, 2, 3, \dots$$

$$b_0 = \frac{1}{2L} \int_c^{c+2L} F(x) dx$$

$$b_n = \frac{1}{L} \int_c^{c+2L} F(x) \cos \frac{n\pi}{L} x dx, \quad n = 1, 2, 3, \dots$$

where c is any real constant.

Fourier Series: Table

Fourier Series in the Finite Interval $(0, L)$

Fourier expansion: $f(x) = \sum_{n=0}^{\infty} A_n \phi_n(x), \quad 0 < x < L$

Expansion coefficients: $A_n = \frac{1}{N_n} \int_0^L f(x) \phi_n(x) dx$

Characteristic-value problem; $\left\{ \begin{array}{l} \frac{d^2 y}{dx^2} + \lambda^2 y = 0 \\ \alpha_1 y(0) + \beta_1 \frac{dy(0)}{dx} = 0 \\ \alpha_2 y(L) + \beta_2 \frac{dy(L)}{dx} = 0 \end{array} \right\}$

Boundary conditions		Characteristic function		Characteristic values λ_n
At $x = 0$	At $x = L$	$\phi_n(x)^{\dagger}$	$N_n = \int_0^L [\phi_n(x)]^2 dx$	are positive roots of ^t
Third kind [†] ($\alpha_1 \neq 0, \beta_1 = 0$)	Third kind [†] ($\alpha_2 \neq 0, \beta_2 \neq 0$)	$\lambda_n \cos \lambda_n x - H_1 \sin \lambda_n x$	$\frac{1}{2} \left\{ (\lambda_n^2 + H_1^2) \left[L + \frac{H_2}{\lambda_n^2 + H_2^2} \right] - H_1 \right\}$	$\tan \lambda L = \frac{\lambda(H_2 - H_1)}{\lambda^2 + H_1 H_2}$
Third kind ($\alpha_1 \neq 0, \beta_1 \neq 0$)	Second kind ($\alpha_2 \neq 0, \beta_2 \neq 0$)	$\cos \lambda_n (L - x)$	$\frac{1}{2\lambda_n} (\lambda_n L + \sin \lambda_n L \cos \lambda_n L)$	$\lambda \tan \lambda L = -H_1$
Third kind ^s ($\alpha_1 \neq 0, \beta_1 \neq 0$)	First kind ^s ($\alpha_2 \neq 0, \beta_2 \neq 0$)	$\sin \lambda_n (L - x)$	$\frac{1}{2\lambda_n} (\lambda_n L + \sin \lambda_n L \cos \lambda_n L)$	$\lambda \tan \lambda L = H_1$
Second kind ($\alpha_1 \neq 0, \beta_1 \neq 0$)	Third kind ($\alpha_2 \neq 0, \beta_2 \neq 0$)	$\cos \lambda_n x$	$\frac{1}{2\lambda_n} (\lambda_n L + \sin \lambda_n L \cos \lambda_n L)$	$\lambda \tan \lambda L = -H_2$
Second kind ($\alpha_1 \neq 0, \beta_1 \neq 0$)	Second kind ($\alpha_2 = 0, \beta_2 \neq 0$)	$\cos \lambda_n x$	$\frac{L}{2}$	$\sin \lambda L = 0$ $\left(\lambda_n = \frac{n\pi}{L}, n = 0, 1, 2, \dots \right)$
Second kind ($\alpha_1 \neq 0, \beta_1 \neq 0$)	First kind ($\alpha_2 \neq 0, \beta_2 \neq 0$)	$\cos \lambda_n x$	$\frac{L}{2}$	$\cos \lambda L = 0$ $\left(\lambda_n = \frac{2n-1}{L} \frac{\pi}{2}, n = 1, 2, 3, \dots \right)$

(Continued)

Fourier Series: Table

Fourier Series in the Finite Interval $(0, L)$

First kind ($\alpha_1 \neq 0, \beta_1 \neq 0$)	Third kind [#] ($\alpha_2 \neq 0, \beta_2 \neq 0$)	$\sin \lambda_n x$	$\frac{1}{2\lambda_n} (\lambda_n L - \sin \lambda_n L \cos \lambda_n L)$	$\lambda \cot \lambda L = -H_2$
First kind ($\alpha_1 \neq 0, \beta_1 \neq 0$)	Second kind ($\alpha_2 \neq 0, \beta_2 \neq 0$)	$\sin \lambda_n x$	$\frac{L}{2}$	$\cos \lambda L = 0$ $\left(\lambda_n = \frac{2n-1}{L} \frac{\pi}{2} \quad n = 1, 2, 3, \dots \right)$
First kind ($\alpha_1 \neq 0, \beta_1 \neq 0$)	First kind ($\alpha_2 \neq 0, \beta_2 \neq 0$)	$\sin \lambda_n x$	$\frac{L}{2}$	$\sin \lambda L = 0$ $\left(\lambda_n = \frac{n\pi}{L} \quad n = 1, 2, 3, \dots \right)$

[†] $H_1 = \alpha_1/\beta_1$ and $H_2 = \alpha_2/\beta_2$.

[‡] $\lambda_0 = 0$ is a characteristic value if $L = 1/H_1 - 1/H_2 > 0$. The corresponding characteristic function is $\phi_0(x) = x - 1/H_1$.

[§] $\lambda_0 = 0$ is a characteristic value if $L = 1/H_1 > 0$. The corresponding characteristic function is $\phi_0(x) = L - x$.

^{||} When $n = 0$, replace $L/2$ by L .

[#] $\lambda_0 = 0$ is a characteristic value if $L = -1/H_2 > 0$. The corresponding characteristic function is $\phi_0(x) = x$.

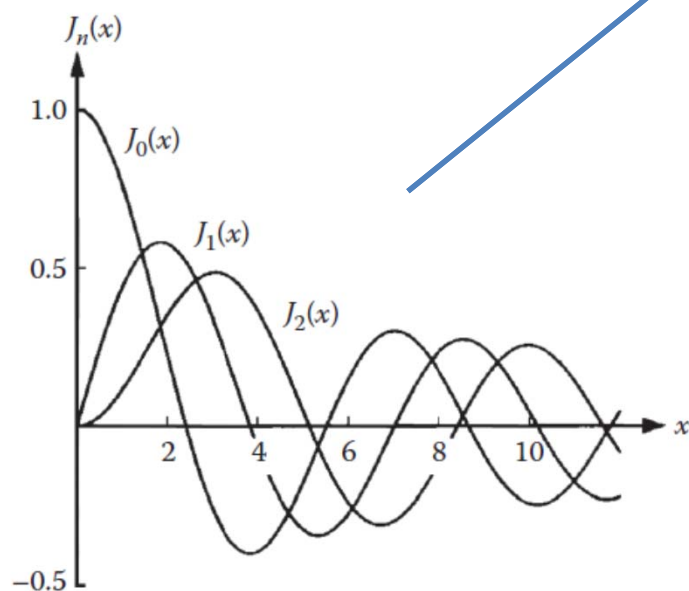
- **Fourier Bessel Series**
- **Fourier Legendre Series**

Bessel Functions

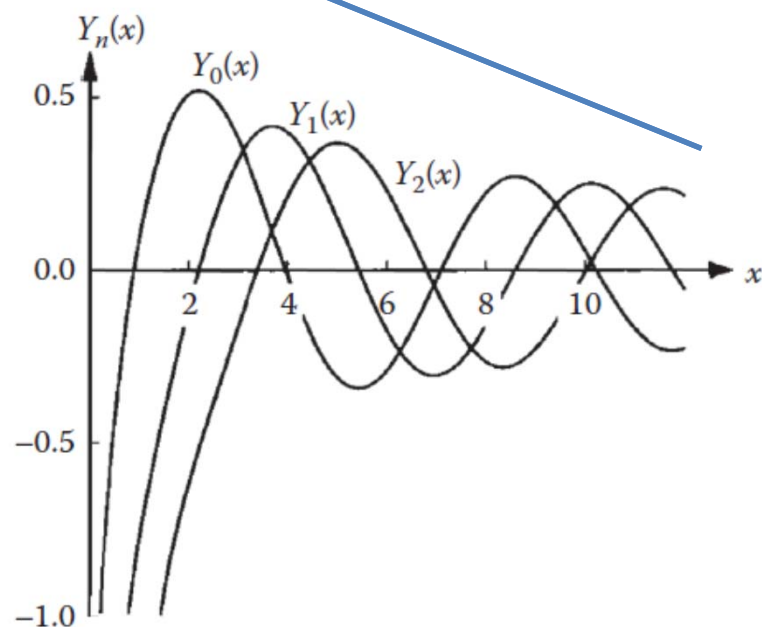
$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (m^2 x^2 - v^2) y = 0$$

Bessel's differential equation of order v , where m is a parameter and v is any real constant.

General solution: $y(x) = C_1 J_v(mx) + C_2 Y_v(mx)$



Bessel functions of the first kind



Bessel functions of the second kind

Modified Bessel Functions

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} - (m^2 x^2 + v^2) y = 0$$

Modified Bessel's differential equation of order v .

General solution: $y(x) = C_1 J_v(mx) + C_2 K_v(mx)$

Numerical Values of $J_n(x)$, $Y_n(x)$, $I_n(x)$, and $K_n(x)$

X	$J_0(x)$	$J_1(x)$	$Y_0(x)$	$Y_1(x)$	$I_0(x)$	$I_1(x)$	$K_0(x)$	$K_1(x)$
0.0	1.0000	0.000	$-\infty$	$-\infty$	1.000	0.000	∞	∞
0.1	0.9975	0.0499	-1.5342	-6.4590	1.0025	0.0501	2.4271	9.8538
0.2	0.9900	0.0995	-1.0811	-3.238	1.0100	0.1005	1.7527	4.7760
0.3	0.9776	0.1483	-0.8073	-2.2931	1.0226	0.1517	1.3725	3.0560
0.4	0.9604	0.1960	-0.6060	-1.7809	1.0404	0.2040	1.1145	2.1844
0.5	0.9385	0.2423	-0.4445	-1.4715	1.0635	2.2579	0.9244	1.6564
0.6	0.9120	0.2867	-0.3085	-1.2604	1.0920	0.3137	0.7775	1.3028
0.7	0.8812	0.3290	-0.1907	-1.1032	1.1263	0.3719	0.6605	1.0503
0.8	0.8463	0.3688	-0.0868	-0.9781	1.1665	0.4329	0.5653	0.8618
0.9	0.8075	0.4059	0.0056	0.8731	1.2130	0.4971	0.4867	0.7165
1.0	0.7652	0.4401	0.0883	-0.7812	1.2661	0.5652	0.4210	0.6019

Fourier Bessel Series

Series expansions in terms of Bessel functions arise most frequently in connection with the following characteristic-value problem (r -direction):

$$r^2 \frac{d^2 R}{dr^2} + r \frac{dR}{dr} + (\lambda^2 r^2 - v^2) R = 0 \qquad \alpha_1 R(a) + \beta_1 \frac{dR(a)}{dr} = 0, \quad \alpha_1^2 + \beta_1^2 > 0$$

This is a special case of the Sturm–Liouville system with:

$$\alpha_2 R(b) + \beta_2 \frac{dR(b)}{dr} = 0, \quad \alpha_2^2 + \beta_2^2 > 0$$

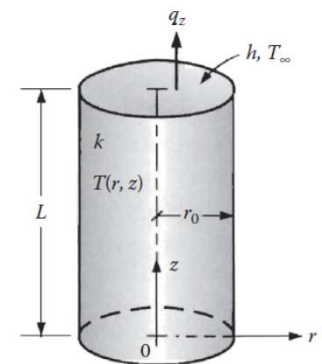
$p(r) = r, \quad q(r) = -\frac{v^2}{r}, \quad \text{and} \quad w(r) = r$ **Compare:** $\frac{d}{dx} \left[p(x) \frac{dy}{dx} \right] + [q(x) + \lambda w(x)] y = 0$
(Divide the DE throughout by r)

Hence, the characteristic functions of this problem form a complete orthogonal set with respect to the weight function r on the interval (a, b) .

For Solid Cylinders (r -direction):

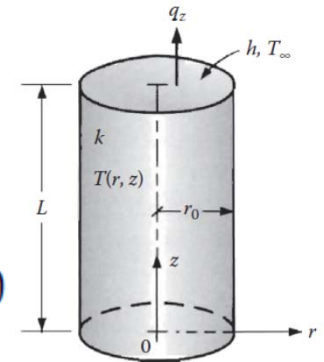
$$r^2 \frac{d^2 R}{dr^2} + r \frac{dR}{dr} + (\lambda^2 r^2 - v^2) R = 0 \qquad R(0) = \text{finite}$$

$$\alpha R(r_0) + \beta \frac{dR(r_0)}{dr} = 0, \quad \alpha^2 + \beta^2 > 0$$



Fourier Bessel Series

$$(A) \begin{cases} r^2 \frac{d^2 R}{dr^2} + r \frac{dR}{dr} + (\lambda^2 r^2 - v^2) R = 0 \\ R(0) = \text{finite} \\ \alpha R(r_0) + \beta \frac{dR(r_0)}{dr} = 0, \quad \alpha^2 + \beta^2 > 0 \end{cases}$$



The general solution of above **Eq. (A)** can be written as:

$$R(r) = AJ_v(\lambda r) + BY_v(\lambda r)$$

The first boundary condition yields $B \equiv 0$. Hence, the characteristic functions are $J_v(\lambda r)$ and the characteristic values λ_n are the roots of the *characteristic-value Eq*:

$$\alpha J_v(\lambda r_0) + \beta \frac{dJ_v(\lambda r_0)}{dr} = 0 \Rightarrow \lambda_n, n = 1, 2, 3, \dots$$

In view of the fact that $p(0) = 0$ and $R(0) = \text{finite}$ (and also $dR(0)/dr = \text{finite}$), the characteristic functions of the system **(A)** form an orthogonal set with respect to the weight function $w(r) = r$ over the interval $(0, r_0)$; that is,

$$\int_0^{r_0} J_v(\lambda_m r) J_v(\lambda_n r) r dr = 0, \quad \lambda_m \neq \lambda_n$$

and the set $\{J_v(\lambda_n r); n = 1, 2, \dots\}$ is a complete orthogonal set.

Thus, we can expand an arbitrary function $f(r)$, which is piecewise differentiable on the interval $(0, r_0)$, in a series of these characteristic functions in the same interval.

Fourier Bessel Series

$$\int_0^{r_0} J_v(\lambda_m r) J_v(\lambda_n r) r dr = 0, \quad \lambda_m \neq \lambda_n$$

$$f(r) = \sum_{n=1}^{\infty} A_n J_v(\lambda_n r), \quad 0 < r < r_0 \quad \text{Fourier-Bessel series of } f(r) \text{ on the interval } (0, r_0).$$

$$A_n = \frac{1}{N_n} \int_0^{r_0} f(r) J_v(\lambda_n r) r dr \quad N_n = \int_0^{r_0} J_v^2(\lambda_n r) r dr$$

$$N_n = \frac{r_0^2}{2\lambda_n^2} \left\{ \left[\lambda_n^2 - \left(\frac{v}{r_0} \right)^2 \right] J_v^2(\lambda_n r_0) + \left[\frac{dJ_v(\lambda_n r_0)}{dr} \right]^2 \right\}$$

Fourier Bessel Series: Table

Fourier–Bessel Series in the Finite Interval $(0, r_0)$

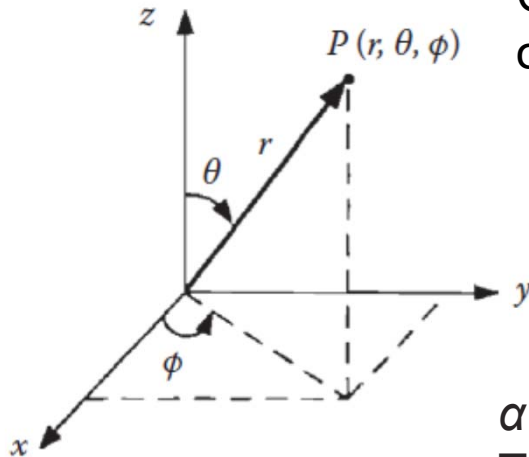
Fourier–Bessel expansion: $f(r) = \sum_{n=1}^{\infty} A_n J_v(\lambda_n r), \quad 0 < r < r_0$	$\left\{ \begin{array}{l} r^2 \frac{d^2 R}{dr^2} + r \frac{dR}{dr} + (\lambda^2 r^2 - v^2) R = 0 \\ R(0) = \text{finite} \\ \alpha R(r_0) + \beta \frac{dR(r_0)}{dr} = 0 \end{array} \right\}$	
Expansion coefficients: $A_n = \frac{1}{N_n} \int_0^{r_0} f(r) J_v(\lambda_n r) r dr$		
Boundary condition at $r = r_0$	$N_n = \int_0^{r_0} J_v^2(\lambda_n r) r dr^{\dagger}$	Characteristic values λ_n are positive roots of [†]
Third kind [‡] ($\alpha \neq 0, \beta \neq 0$)	$\frac{r_0^2}{2} \left[1 + \frac{1}{\lambda_n^2} \left(H^2 - \frac{v^2}{r_0^2} \right) \right] J_v^2(\lambda_n r_0)$	$H J_v(\lambda r_0) + \frac{dJ_v(\lambda r_0)}{dr} = 0$
Second kind ($\alpha \neq 0, \beta = 0$)	$\frac{r_0^2}{2} \left[1 - \left(\frac{v}{\lambda_n r_0} \right)^2 \right] J_v^2(\lambda_n r_0)$	$\frac{dJ_v(\lambda r_0)}{dr} = 0^{\S}$
First kind ($\alpha = 0, \beta \neq 0$)	$\frac{r_0^2}{2} J_{v+1}^2(\lambda_n r_0)$	$J_v(\lambda r_0) = 0$

[†] $H = \alpha/\beta$.

[‡] When $v > 0$ and $r_0 = -v/H > 0$, $\lambda_0 = 0$ is a characteristic value for this case with the corresponding characteristic function $\phi_0(r) = r^v$.

[§] When $v = 0$, $\lambda_0 = 0$ is also a characteristic value for this case with the corresponding characteristic function $\phi_0(r) = 1$.

Legendre Polynomials



Consider the following linear and homogeneous second-order ordinary differential equation with variable coefficients

$$(1-x^2)\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + \alpha(\alpha+1)y = 0$$
$$\Rightarrow \frac{d}{dx}\left[(1-x^2)\frac{dy}{dx}\right] + \alpha(\alpha+1)y = 0$$

$\alpha = \text{real, non-negative.}$

This is *Legendre's differential equation*

Its two linearly independent solutions are called

- *Legendre functions of the first kind*
- *Legendre function of the second kind*

When $\alpha = n$, where n is zero or a positive integer, the Legendre function of the first kind becomes a polynomial in x of degree n , whereas the second kind can be expressed as an infinite series. The general solution of Legendre's equation for $n = 0, 1, 2, \dots$ can be written as:

$$y(x) = C_1 P_n(x) + C_2 Q_n(x)$$

where $P_n(x)$ are polynomials called *Legendre polynomials of degree n* , and $Q_n(x)$ are the *Legendre functions of the second kind* (which converges when $|x| < 1$, but diverges as $x \rightarrow \pm 1$).

Legendre Polynomials

$$(1-x^2) \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} + n(n+1)y = 0$$

$$\Rightarrow \frac{d}{dx} \left[(1-x^2) \frac{dy}{dx} \right] + n(n+1)y = 0$$

substitute $x = \cos \theta$

By chain rule:

$$\frac{d}{dx} = \frac{d\theta}{dx} \frac{d}{d\theta} = -\frac{1}{\sin \theta} \frac{d}{d\theta}$$

$$\text{Also, } 1-x^2 = 1-\cos^2 \theta = \sin^2 \theta$$

Now Legendre equation takes the form,

$$-\frac{1}{\sin \theta} \frac{d}{d\theta} \left[\sin^2 \theta \left(-\frac{1}{\sin \theta} \right) \frac{dy}{d\theta} \right] + n(n+1)y = 0$$

$$\Rightarrow \frac{1}{\sin \theta} \frac{d}{d\theta} \left[\sin \theta \frac{dy}{d\theta} \right] + n(n+1)y = 0$$

$$\text{Equivalently, } \frac{d^2 y}{d\theta^2} + \cot \theta \frac{dy}{d\theta} + n(n+1)y = 0$$

Legendre Polynomials

$$\frac{d}{dx} \left[(1-x^2) \frac{dy}{dx} \right] + \alpha(\alpha+1)y = 0$$

$\alpha = \text{real, non-negative.}$

This is *Legendre's differential equation*

The substitution of $x = \cos \theta$ transforms Legendre's differential equation as:

$$\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{dy}{d\theta} \right) + n(n+1)y = 0$$

where $\alpha = n$, where n is
zero or a positive integer.

$$\Rightarrow \frac{d^2 y}{d\theta^2} + \cot \theta \frac{dy}{d\theta} + n(n+1)y = 0$$

Solution: $y(\theta) = C_1 P_n(\cos \theta) + C_2 Q_n(\cos \theta)$

Note that $Q_n(\cos \theta)$ is not finite when $\cos \theta = \pm 1$, that is, when $\theta = k\pi$, $k = 0, \pm 1, \pm 2, \dots$

And, $P_n(\cos \theta)$ is merely a polynomial of degree n in $\cos \theta$. In particular,

$$P_0(\cos \theta) = 1, \quad P_1(\cos \theta) = \cos \theta \quad P_2(\cos \theta) = \frac{1}{2}(3\cos^3 \theta - 1) = \frac{1}{4}(3\cos 2\theta + 1)$$

Fourier Legendre Series

Compare Legendre DE with: $\frac{d}{dx}\left[p(x)\frac{dy}{dx}\right] + [q(x) + \lambda w(x)]y = 0$

$$\frac{d}{dx}\left[(1-x^2)\frac{dy}{dx}\right] + \alpha(\alpha+1)y = 0 \quad \text{Eq. (A)}$$

$$p(x) = 1 - x^2, \quad q(x) = 0, \quad w(x) = 1, \quad \lambda = \alpha(\alpha + 1)$$

Since the function $p(x)$ vanishes at $x = \pm 1$, any two distinct solutions of Eq. (A) which are finite and have finite first derivatives at $x = \pm 1$ will be orthogonal with respect to the weight function $w(x) = 1$ on the interval $(-1, 1)$. Therefore, no specific boundary conditions are needed for Eq. (A) to lead to a Sturm–Liouville type characteristic-value problem over the interval $(-1, 1)$.

Equation (A) possesses solutions that are finite at $x = \pm 1$ only if α is zero or a positive integer. Therefore, $\alpha = n$, where $n = 0, 1, 2, \dots$

The corresponding solutions that are finite when $x = \pm 1$ are proportional to the Legendre polynomials $P_n(x)$. Thus,

$$\int_{-1}^1 P_m(x)P_n(x)dx = 0 \quad m \neq n$$

The set $\{P_n(x); n = 0, 1, 2, \dots\}$ is a complete orthogonal set with respect to the weight function *unity* over the interval $(-1, 1)$.

Fourier Legendre Series

$$\int_{-1}^1 P_m(x)P_n(x)dx = 0 \quad m \neq n$$

The set $\{P_n(x); n = 0, 1, 2, \dots\}$ is a complete orthogonal set with respect to the weight function *unity* over the interval $(-1, 1)$.

Therefore any piecewise differentiable function $f(x)$ can be represented in the interval $(-1, 1)$ by a series of the Legendre polynomials as

$$\begin{aligned} f(x) &= A_0P_0(x) + A_1P_1(x) + A_2P_2(x) + \dots \\ &= \sum_{n=0}^{\infty} A_nP_n(x), \quad -1 < x < 1 \end{aligned}$$

which is called the *Fourier–Legendre series expansion* of $f(x)$ on the interval $(-1, 1)$. The expansion coefficients A_n are given by

$$A_n = \frac{\int_{-1}^1 f(x)P_n(x)dx}{\int_{-1}^1 [P_n(x)]^2 dx}$$

It can be shown that:

$$\int_{-1}^1 [P_n(x)]^2 dx = \frac{2}{2n+1} \quad \Rightarrow \quad A_n = \frac{2n+1}{2} \int_{-1}^1 f(x)P_n(x)dx$$

Thank You